Aishwarya Shyam

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PROFILE

Software Engineer with 4 years of experience in software development and a Master's in Computer Science. Skilled in JavaScript, Python, and AWS, with expertise in designing RESTful APIs, developing reusable UI components, optimising application performance, and building intelligent automation systems powered by modern AI technologies. Strong track record of delivering end-to-end features in Agile teams using modern CI/CD.

SKILLS

AI: GenAI, LLMs, Prompt Engineering, LLM APIs (OpenAI), LiteLLM, Agentic AI, RAG, Vector Databases, MCP, AI-Assisted Development (Codex)
Backend: Python, Flask, Node.js, REST APIs, Microservices, Redis, Kafka, SQL, PostgreSQL, MongoDB
Frontend: JavaScript, TypeScript, React.js, Next.js, Angular, HTML, CSS, Tailwind CSS, UI/UX fundamentals
Cloud & Infra: AWS (EC2, S3, IAM, Lambda), Serverless Architecture, Docker, Kubernetes, CI/CD, Jenkins, Linux, NGINX

WORK EXPERIENCE

Open Science Tools Nottingham, UK
Software Engineer Jul 2024 – Apr, 2026

- Designed and maintained **10+ REST APIs** using **Python, Flask,** and Gunicorn to support user onboarding, account management, and internal platform workflows across a production platform serving **50,000+ users**.
- **Led end-to-end deployment** of a secure, self-hosted Pavlovia instance for a client, managing **infrastructure**, GitLab installation, **SSL/TLS configuration**, server hardening, and ongoing production support.
- Automated GitLab domain whitelist updates using a **Python** script and GitLab **CI/CD** environment variables, ensuring consistent, error-free updates without manual intervention and **reducing configuration effort by 80%.**
- Owned and built an access management portal, integrating **Stripe Payment Intents** and controlled account creation via the **GitLab API**, reducing admin overhead during onboarding workflows by **60%.**
- Built a **backend log processing system** to transform production logs into structured JSON, improving **debugging.**
- Implemented unit and integration tests using **Pytest**, and automated **Playwright**-based end-to-end tests for Pavlovia.
- Integrated Google and Microsoft as OAuth 2.0 providers to enable secure single sign-on (**SSO**) for Pavlovia users.
- Collaborated with **stakeholders** and **clients** to scope features and assess feasibility.

Mphasis Bangalore, IN
Associate Solution Developer Jul 2021 – Dec 2022

- Demonstrated proficiency in core web development technologies, including **JavaScript/TypeScript, HTML** and **CSS**, while collaborating within the **Blockchain Centre of Excellence** team.
- **Led the end-to-end development of 3 Web3.0 applications** using the MERN stack (MongoDB, Express.js, **React/Next.js, Node.js**), building decentralised solutions such as an NFT Marketplace.
- Designed and optimised RESTful APIs using **Node.js**, improving response times by 50%.
- Implemented Corda flows and contract states in **Java** to manage peer-to-peer distributed transactions.
- Collaborated on **5+ proof-of-concepts**, integrating blockchain features and deploying applications as **Docker** containers on **AWS** using CI/CD pipelines. Implemented **unit and integration tests** using **Jest** to improve code quality.

Amazon Bangalore, IN
Transaction Risk Investigator Nov 2020 – Jul 2021

- Performed large-scale transaction risk analysis using **SQL** and internal fraud detection platforms, identifying behavioural anomalies and compliance risks across 500+ customer accounts.

EDUCATION

University of Strathclyde Glasgow, UK
Masters in Advanced Computer Science Jan 2023 – Jan 2024

Visvesvaraya Technological University Bangalore, IN
Bachelor of Engineering (Information Science) Jul 2016 – Jul 2020

PROJECTS

Job Search Copilot | *Python, LiteLLM, OpenAI API, Playwright, Supabase, GitHub Actions* July 2026 – Ongoing
Building an AI-powered job search copilot that automates job discovery, resume tailoring, outreach drafting and application tracking using LLMs. Designed a provider-agnostic AI architecture using LiteLLM, GitHub Actions, Supabase and Playwright for scalable agent workflows.

ReframeAI | *Python, Flask, OpenRouter API, HTML, CSS* May 2026
Developed a full-stack AI-powered reflection application that analyses user input for cognitive distortions and generates personalised CBT reframes.

GlucoCare | *React.js, Node.js, Express.js, MongoDB, JWT* Jul 2023 – Aug 2023
Developed a full-stack diabetes platform with health tracking, personalised recommendations, discussion forums, and secure JWT authentication.

University Distributed Information Platform | *MySQL, XML, SPARQL, XQuery, Protégé* Feb 2023 – Apr 2023
Designed and implemented an enterprise-scale distributed information platform integrating student, course, and departmental data using relational, XML, and semantic data models, with SQL, XQuery, and SPARQL for cross-domain data access.